

TECHNICAL MANUSCRIPT

The Current Proof Route for Shared-Probe HSS Approximation

The synchronized learner, proved components,
and the outstanding approximation estimate

Research snapshot: 20 September 2026

Status. Independent fixed-space refitting is proved. The general adaptive learning estimate and a computationally justified final rank reduction remain outstanding.

Prepared from the supplied proof manuscript and the subsequent proof, audit, and numerical records.

Real matrices. Fixed inputs. Exact arithmetic in theorem statements.
Squared Frobenius error and explicit conditioning conventions.

Abstract

This manuscript states the current route toward an $O(k)$ -query, rank- k HSS approximation algorithm with expected squared Frobenius error $O(L) \text{OPT}_k(A)$. It is a proof-status manuscript, not a completed proof of the conjecture. The fixed-space refitting component is established: independent probes yield a depth-uniform factor 1607 with $8s$ queries for prescribed nested bases of rank at most s .

The current learning candidate uses synchronized row and column unions, a full raw-range branch, canonical completion, and original-data cut commutators. The exact-input capture theorem, including an enlarged exact-rank endpoint, is separated from residual-relative stability. A fixed-comparator right separator gives an exact signed identity for new physical loss. Its unlifted source has a total budget at most $24qL\|A - B\|_F^2$. The unresolved step is a joint estimate after the adaptive left lift, retaining the correlation between the source and inherited child error and accounting for any invisible component.

Restricted regression-tree theorems, guard completion, and noncentral projector identities provide rigorously stated auxiliary tools. Their conditioning and independence hypotheses are retained. Numerical checks of the actual synchronized recursion and the signed identity are summarized as diagnostics, with their sample sizes and scope. Conditional assembly of the full approximation chain identifies two distinct remaining tasks: the general basis-quality estimate and a computationally justified reduction to final rank k . Exact offline HSS minimization provides a query-only reduction, not an efficient algorithm by itself.

How to read this proof record

This document is frozen at the user’s instruction to stop research on 20 September 2026. It consolidates existing results from the supplied manuscript and subsequent local proof notes. No new experiment or proof search was undertaken for this consolidation. “Audited” means reviewed in the recorded internal mathematical audit, not externally peer reviewed.

The public target is the query-count improvement raised in Amsel et al. [1]. Its proved benchmark has $O(kL)$ matrix–vector queries and an $O(L)$ expected squared-error factor. The aim here is to retain the latter factor with only $O(k)$ queries. Results for exact HSS inputs, larger output rank, a prescribed learned coefficient space, or unrestricted offline optimization do not alone establish that target.

The active candidate in this record is the *synchronized* union construction. Earlier full-range and separately trained two-view candidates have distinct counterexamples. They are documented in the companion manuscript, *Obstructions in Shared-Probe HSS Approximation*, and must not be conflated with the active algorithm. The source filenames printed beside results identify the proof notes and audits used in this consolidation.

The main logical chain is

$$\begin{aligned} \text{shared-probe learning} &\longrightarrow \text{nested coefficient space} \\ &\longrightarrow \text{independent refit} \longrightarrow \text{rank-}k \text{ output.} \end{aligned}$$

The middle refit implication is proved. The learning estimate remains open, and the final computational implication must be justified separately. Restricted-model results are presented with their original hypotheses; they are not substituted for the missing general learner theorem.

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The current route and the theorem still required

1.1 Target, conventions, and status

Let $\mathcal{T}$ be a fixed balanced binary partition tree of $\{1, \dots, N\}$. The target class uses physical leaves of size at most $2k$. For a nonroot node I , write I^c for its complement and P_I for the diagonal coordinate projector onto I . An HSS matrix of rank at most k has outgoing and incoming exterior-cut ranks at most k at every nonroot node, with nested row and column bases on this tree. Equivalently, these cut ranges admit the usual nested HSS representation. Physical-leaf diagonal blocks are unrestricted. Throughout this chapter,

$$e_\star = \text{OPT}_k(A) := \inf_{B \in \text{HSS}_k(\mathcal{T})} \|A - B\|_F^2$$

is a *squared* Frobenius error. Let $L \geq 1$ bound the number of nonroot levels. Its order is $\log(N/k)$; changing a bounded number of bottom levels changes this convention only by an absolute additive constant.

The desired theorem is an algorithm which, for every fixed real matrix A , returns $\hat{B} \in \text{HSS}_k(\mathcal{T})$ and satisfies

$$\mathbb{E} \|A - \hat{B}\|_F^2 \leq CL \text{OPT}_k(A), \quad (1.1)$$

using $O(k)$ total products with A and A^T and polynomial additional work. The constant must be independent of dimension, depth, input conditioning, and the size of the best-approximation error. A method that is exact only on HSS inputs, a method returning rank $4k$, and a query-efficient method with an unimplemented global optimization do not individually establish this target.

The established comparison point is the $O(kL)$ -query, rank- k , $O(L)$ expected squared-error guarantee of Amsel et al. [1]. The reused-sketch algorithm of Levitt and Martinsson [2] has a linear query count, but its stated analysis does not supply the required arbitrary-input approximation guarantee for the adaptive coarser-level spaces. These are the specific primary statements relevant here; this manuscript makes no claim that a bounded literature check proves absence of every possible later result. The versions and scope checks are recorded in `latest_primary_sources.md`.

The current route separates three issues. A learned nested space must have small *physical coefficient-space approximation error*; an independent query batch must recover its coefficients stably; and the result must be reduced to rank k on the original target tree with the required computational cost. The coefficient-recovery step is proved. Exact capture and several local identities for a new synchronized learner are proved. Its general noisy physical-stability estimate and an efficient same-rank final reduction are not proved.

1.1.1 Three learners which must not be identified

The following distinctions are part of the mathematical statements.

1. The *separately trained maintained union* constructs an annihilated hierarchy and a raw hierarchy independently, each with retained width $2k$, then takes their physical unions. Its usual parameters are query width $16k$ per stream and leaves of size $4k$. The final union has width at most $4k$. Unequal intermediate templates are intrinsic to that construction.
2. The *completed synchronized learner*, defined below, feeds each current child union into both views of the next merge. Its per-view truncation width is $r = 2k$, retained width is $R = 4k$, query width is $q = 32k$ per stream, and physical leaves have size at most $8k$. It also keeps an entire raw fit whenever that fit has rank at most R . These are substantive algorithmic changes.
3. The *fixed-source full-fit learners* studied later retain an entire matrix-target fit and use either specified prefix or fixed guard completion. Their assumptions include independent Gaussian source blocks on fixed disjoint supports. They are restricted local models, rather than alternative names for either HSS learner.

An obstruction for one of these rules is not automatically an obstruction for the others. Conversely, a theorem for a restricted source model cannot be imported merely because its local regression formula resembles an HSS merge.

The synchronized learner uses a coarsening of the original target tree: its $8k$ physical leaves combine a bounded number of $2k$ target leaves. Every comparator on the original tree remains a valid comparator on this coarsened tree. The converse is unnecessary and generally false. In particular, arbitrary dense blocks on the coarser physical leaves must not be silently declared rank-compatible on a finer target tree.

1.2 The independent fixed-space refit

Fix nested orthonormal row and column bases of width s , allowing endpoint dimensions to be capped by their physical sizes. Use physical leaves of size at most $2s$. Let $\mathcal{S}$ be their linear coefficient space: sibling interactions have the prescribed row and column spaces, physical-leaf diagonal blocks are arbitrary, and the root retains its full child coefficient block. For fixed bases this is a linear space of matrices, not the nonlinear class of all HSS matrices of rank s .

For clarity, if a, b are siblings, the orthogonal projection of a matrix A onto the corresponding coefficient block is

$$U_a U_a^T A[a, b] V_b V_b^T.$$

Together with the reversed sibling blocks and unchanged leaf diagonal blocks, these disjoint blocks specify the orthogonal projection onto $\mathcal{S}$. Its error is denoted $\text{dist}_F(A, \mathcal{S})^2$. Computing this projection from the dense matrix is a useful diagnostic; it is not the query-only refit algorithm.

Theorem 1.1 (Independent fixed-space refit). *Suppose the nested bases defining $\mathcal{S}$ are fixed independently of two new standard Gaussian query streams. There is a sibling-orthogonal linear coefficient refit using width $4s$ in each stream, hence $8s$ forward and transpose products in total, whose output $C \in \mathcal{S}$ satisfies*

$$\mathbb{E}[\|A - C\|_F^2 \mid \mathcal{S}] \leq 1607 \text{dist}_F(A, \mathcal{S})^2. \quad (1.2)$$

The constant is independent of tree depth and of the geometry of the fixed bases.

The refit uses the known nested bases and the new products to perform linear sibling-orthogonal coefficient regressions and their prescribed top-down corrections. It does not learn new bases. Its detailed sourcewise proof and independent mathematical review are recorded in `audit_refit.md` and `hss_complete_proof_chain.tex`.

Here is the proof structure relevant to subsequent applications. After conditioning on the bases, retained and discarded query coordinates admit the fixed-frame Gaussian quotient and Haar decompositions. The physical residual splits into orthogonal source sectors: interior discarded components, sibling interaction components, and their source-to-root terminal outputs. Centering the corrections in the appropriate top-down order gives orthogonality between same-sided source contributions. The outputs of the two query streams are not declared independent; their combination is bounded in L^2 .

The retained coefficient recurrence cancels the intermediate feedback terms before any norm bound is taken. The remaining terms are the transported retained coefficient error and the direct discarded-source contribution; the detailed notation and cancellation are given in the fixed-space refit proof in `hss_complete_proof_chain.tex`, as checked by `audit_refit.md`. This cancellation, rather than a sum of separately bounded feedback terms, yields the retained-output gain. At query width $4s$, the quotient ratio is $s/(q - 2s) = 1/2$. The sourcewise predictable-energy bound, the terminal source-product estimate, and the frozen-stream feedback bound are summable. The audited assembly, with its specified slack parameters, gives the constant 1607. This explanation identifies the proof's independence interface: conditioning on the learned bases is permitted only because the refit streams are genuinely new.

For the synchronized learner, take $s = R = 4k$. Its physical leaves have size $8k = 2s$, so this theorem applies after training. Training costs $64k$ products and this refit costs $32k$, for a conditional total of $96k$. If a basis theorem established

$$\mathbb{E} \text{dist}_F(A, \mathcal{S})^2 \leq C_b L \text{OPT}_k(A), \quad (1.3)$$

the refit would give the same order of error in the learned rank- $4k$ coefficient space. Equation (1.3) is the unproved general-input interface, not a consequence of Theorem 1.1.

1.2.1 What an exact final projection would establish

Suppose, as a mathematical operation, one chooses a best HSS-rank- k approximation $\hat{B}$ to C on the original target tree. For a best comparator $B_\star$, or by a limiting argument with approximate minimizers,

$$\|A - \hat{B}\|_F \leq \|A - C\|_F + \|C - \hat{B}\|_F \quad (1.4)$$

$$\leq 2\|A - C\|_F + \sqrt{\text{OPT}_k(A)}. \quad (1.5)$$

Consequently (1.3) and the refit imply

$$\mathbb{E}\|A - \hat{B}\|_F^2 \leq (2\sqrt{1607C_b L} + 1)^2 \text{OPT}_k(A).$$

This is a *query-only reduction*. It does not provide a polynomial-time implementation of best HSS projection or a verified constant-factor same-rank recompression routine. An additional depth-dependent squared-error factor in a recompression theorem would generally produce an extra factor of L here. Thus even completion of the basis theorem would leave the polynomial-work requirement to be settled by an appropriate efficient reduction.

A completed synchronized learner and exact capture

2.1 A query-only rule with all rank branches specified

Set

$$r = 2k, \quad R = 4k, \quad q = 32k.$$

Draw independent standard Gaussian matrices $\Psi, \Omega \in \mathbb{R}^{N \times q}$ once, and obtain $Y = A\Omega$ and $Z = A^T\Psi$. Every node uses these same original arrays. Its accessible response is

$$J_I = \Psi_I^T Y_I - Z_I^T \Omega_I = \Psi^T [P_I, A] \Omega \in \mathbb{R}^{q \times q}. \quad (2.1)$$

No Gaussian design is resampled at an internal node.

Write N_D for a matrix with orthonormal columns spanning the right kernel of D . Using this basis instead of the kernel projector changes only the right coordinates of a nullified sample, not its positive left singular spaces or the selected physical ranges. At a physical leaf I , form the row and column samples

$$Y_I N_{\Omega_I}, \quad Z_I N_{\Psi_I}.$$

If a sample has rank at most R , keep its full positive left singular range. Otherwise keep its leading r positive left singular directions. Complete in physical coordinate order to dimension $\min(R, |I|)$. A zero sample therefore receives the specified fixed coordinate prefix. This full-range branch is part of the algorithm; it is not a proof convention.

At an internal nonroot node with children a, b , both views use the same completed child unions:

$$W = \text{diag}(U_a, U_b), \quad W_V = \text{diag}(V_a, V_b),$$

where disjoint child supports are understood. Their widths are at most $2R = 8k$. In row coordinates put

$$G = W^T \Psi_I, \quad H = W_V^T \Omega_I, \quad L_G = (G^\dagger)^T, \quad \Pi_G = G^\dagger G, \quad Q_G = L_G G^T. \quad (2.2)$$

Here Π_G acts in query space and Q_G acts in physical-template coordinates. Both are orthogonal projectors; these definitions remain valid for singular G . Form

$$M_0 = \Pi_G J_I, \quad M_1 = \Pi_G J_I N_H.$$

If $\text{rank } M_0 \leq R$, the mandatory local space is $\text{range}(L_G M_0)$. Otherwise, take the leading r positive left singular directions of each M_0, M_1 , lift them by L_G , and take their union. This mandatory union has dimension at most R .

There is one common completion, performed *after* this union. Starting with the mandatory space, scan

$$Q_G e_1, \dots, Q_G e_m, \quad (I - Q_G) e_1, \dots, (I - Q_G) e_m,$$

adjoining each nonzero orthogonalized candidate until dimension $\min(R, m)$ is reached. A fixed projector-based coordinate convention chooses the final orthonormal frame U of this space, and

the physical row basis is WU . The two views are not padded separately. The local projector $P = UU^T$ obeys

$$PQ_G = Q_G P. \quad (2.3)$$

Indeed the mandatory directions are visible, and every additional direction is taken within either the visible subspace or its orthogonal kernel. When G has full row rank, this is ordinary coordinate completion.

The column rule is the transpose version, using $-J_I^T$ and exchanging G, H . At the global root there is no exterior-cut selection: the full child coefficient block is retained. All operations are local linear algebra on the original products and stored bases. This construction, its exact theorem, and an independent proof audit are recorded in `synchronized_union_exact_route.md` and `synchronized_union_exact_audit.md`.

2.2 Exact and rank-slack inputs

Theorem 2.1 (Exact capture with rank slack). *Fix an input A for which every outgoing and incoming nonroot exterior cut on the learner tree has rank at most $2k$. Almost surely, all reached internal predesigns have full row rank, and the synchronized row and column spaces contain the corresponding true cut ranges. This includes zero cuts, deficient ranks, and physical endpoint caps. In particular A belongs to the learned coefficient space. Every exact HSS-rank- k input satisfies these hypotheses.*

The full-row-rank conclusion is a theorem on this exact rank-slack family. It does not say that learned designs are Gaussian, and it is not an arbitrary-noisy-input smallest-singular-value estimate.

Proof outline with the conditioning interface. Let $a_I, b_I \leq 2k$ be the outgoing and incoming cut ranks. For every probe realization,

$$\text{rank } J_I \leq a_I + b_I \leq 4k = R.$$

Thus the rule collapses to the full-row branch before any conditioning. The row recursion no longer depends on the opposite learned hierarchy.

Factor the outgoing block as $C_I F_I^T$, with F_I an orthonormal exterior factor, and write the incoming transpose block as $V_I K_I$, with V_I orthonormal on I . Fix Ω on the probability-one event on which its finitely many true-cut anchors and leaf nullified samples have their required ranks. The disjoint-support anchors

$$\Gamma_I = F_I^T \Omega_{I^c}, \quad X_I = V_I^T \Omega_I$$

admit a separator T_I with $\Gamma_I T_I = I$ and $X_I T_I = 0$. At a leaf, $q - |I| \geq 24k$ is more than sufficient for the outgoing rank. The leaf row space therefore consists of the true outgoing cut range and a fixed canonical completion under this conditioning.

For the induction, choose deterministic orthonormal true frames C_I on the outgoing range and Z_I on the exterior incoming range, and put

$$h_I = \Psi_I^T C_I, \quad e_I = \Psi_{I^c}^T Z_I, \quad \mathcal{H}_I = \text{range}(h_I), \quad \mathcal{K}_I = \text{range}[h_I, e_I].$$

If $\mathcal{S}_I$ is the query image of the retained row space and its dimension is $d_I \leq R$, the induction maintains injectivity, true-cut containment, and

$$\mathcal{S}_I \cap \mathcal{K}_I = \mathcal{H}_I.$$

The exterior incoming range for child a is contained in the sum of the sibling outgoing range and the parent's exterior incoming range. Complete these fixed physical frames orthogonally. The newly exposed Gaussian anchor is independent of the child's old physical block and old exterior anchor. The rank inequalities $d_a + a_b + b_I \leq 8k$ and their symmetric versions preserve the required child intersections after these anchors are exposed.

Condition now on Ω, h_a, h_b, e_I , rather than on the entire old query designs. The remaining child Gaussian blocks are independent. Rotations fixing $\text{range}[h_a, h_b, e_I]$ preserve the collapsed full-fit rule and its canonical completion. The two residual child query subspaces are therefore independent Haar Grassmann subspaces in the remaining query coordinates. Their available dimensions, and

$$d_a + d_b + b_I \leq 10k < q,$$

give disjointness and injectivity of the stacked parent design almost surely. The corresponding intersection invariant is preserved.

The child containment implies $C_I \in \text{range } W$. Multiplication by the fixed separator gives

$$L_G J_I T_I = L_G G^T W^T C_I = W^T C_I,$$

where injectivity was just established. The full-raw branch retains this range, proving parent capture. The transpose assertion follows by reversing the roles of the two streams and repeating the argument. One does not condition on both learned hierarchies and then declare the resulting matrices independent Gaussian. $\square$

2.3 The exact clean-parent statement

Fix $A = B + E$ with a fixed HSS-rank- k comparator B . Suppose at a particular parent that the common row template W contains the comparator outgoing cut C , and that $J_I(E) = 0$. A fixed-comparator separator as constructed in the next chapter gives

$$L_G J_I(B) T = Q_G W^T C.$$

Moreover $\text{rank } J_I(B) \leq 2k$, so the entire raw lift is retained. Consequently

$$(I - W P W^T) C = (I - W P W^T) W (I - Q_G) W^T C. \quad (2.4)$$

Only comparator directions invisible to the common design can be lost. If that design is injective, the parent captures the comparator cut exactly, without a hypothesis on containment in the learned opposite frame. This conclusion is a local deterministic statement. It does not identify the actual feedback trajectory with the trajectory of a separately trained two-view hierarchy.

The signed source identity and the remaining physical budget

3.1 A separator determined by the fixed comparator

Fix $A = B + E$, where $B \in \text{HSS}_k(\mathcal{T})$ is deterministic. At a node I , factor

$$B[I, I^c] = CF^T, \quad B[I^c, I]^T = VK,$$

with F, V orthonormal, of widths $a, b \leq k$ respectively. Extend F by zero on I and V by zero outside I . Set

$$\Gamma = F^T \Omega, \quad X = V^T \Omega, \quad \Theta = K \Psi_{I^c}.$$

The comparator commutator is

$$J_I(B) = \Psi_I^T C \Gamma - \Theta^T X. \quad (3.1)$$

The two rows of notation in this formula concern fixed true cut factors, not learned bases. If N_X spans $\ker X$, define

$$T = N_X (\Gamma N_X)^\dagger. \quad (3.2)$$

Almost surely $\Gamma T = I_a$ and $XT = 0$. The separator depends on the fixed comparator and Ω , and is independent of Ψ . It does not depend on the learned annihilator H . This distinction is what makes its fixed-source moments available.

For the local template, decompose

$$C = WC_c + C_\perp, \quad C_c = W^T C, \quad C_\perp = (I - WW^T)C.$$

Then, even for singular G ,

$$L_G J_I(B) T = Q_G C_c + L_G \Psi_I^T C_\perp. \quad (3.3)$$

The inherited defect $C_\perp$ is an adaptive physical matrix; it must not be treated as fixed when applying Gaussian moment formulas.

3.2 A raw tail with an unlifted budget

Let U be the retained local coordinate frame and $P = UU^T$. Define

$$R_0 = (\Pi_G - \text{Proj}(G^T U)) J_I, \quad (3.4)$$

where $\text{Proj}(K) = KK^\dagger$ projects onto a column range. The expression $G^T U$, rather than GU , is essential. The two query projectors are nested, so their difference is an orthogonal projector.

In the high-rank branch, the retained union contains the leading $2k$ raw directions after lifting, and mapping them back by G^T recovers those query directions. Since $\text{rank } \Pi_G J_I(B) \leq 2k$, the Eckart–Young comparison gives

$$\|R_0\|_F \leq \|J_I(E)\|_F. \quad (3.5)$$

In the full-raw branch $R_0 = 0$, so the same inequality holds. No Gaussian independence is used in this pathwise statement.

For a fixed matrix M and two independent Gaussian streams,

$$\mathbb{E}\|\Psi^T M \Omega\|_F^2 = q^2 \|M\|_F^2.$$

At any one partition level, $\sum_I \|[P_I, E]\|_F^2 \leq 2\|E\|_F^2$. Summing over the nonroot levels yields

$$\sum_I \mathbb{E}\|R_{0,I}\|_F^2 \leq 2q^2 L \|E\|_F^2. \quad (3.6)$$

This is an unlifted query-space budget. It has not yet paid for the adaptive dual lift.

3.2.1 The stronger separator-weighted source budget

Let $P_\perp = I - FF^T - VV^T$, with the globally embedded factors above. For every fixed M ,

$$\mathbb{E}\|M \Omega T\|_F^2 = \|MF\|_F^2 + \frac{a}{q - a - b - 1} \|MP_\perp\|_F^2. \quad (3.7)$$

To see this, decompose Ω along the mutually orthogonal physical ranges $F, V, P_\perp$. The first term is fixed by $\Gamma T = I$, the second disappears by $XT = 0$, and the remaining Gaussian block is independent of the two anchors. The inverse-Wishart trace identity for ΓN_X gives the displayed coefficient. Independence of Ψ then multiplies this expression by q when Ψ^T is put on the left. Since $a, b \leq k$ and $q = 32k$, taking $M = [P_I, E]$ gives

$$\mathbb{E}\|J_I(E)T\|_F^2 \leq q\|[P_I, E]\|_F^2. \quad (3.8)$$

It is false in general that the raw-tail inequality can simply be right-weighted pathwise. For example, a leading-one truncation of $\text{diag}(2, 1)$, with comparator $\text{diag}(0, 1)$, residual $\text{diag}(2, 0)$, and $T = e_2$, has weighted tail norm one and weighted residual norm zero. The required tail estimate instead uses moments. A fixed net and the directional inverse-chi-square law give

$$(\mathbb{E}\|T\|_{\text{op}}^4)^{1/2} \leq \frac{2 \cdot 9^{1/3}}{24k + 1}.$$

One way to track the dimensions is to set $d = q - b$, $\nu = d - a + 1$, and use order $3a$ for a net of size at most 9^a . The inverse-chi-square moment is bounded by $1/(\nu - 6a)$ at this order, and the net comparison contributes its factor two. The case $a = 0$ is vacuous.

For the other factor, direct Gaussian fourth moments give

$$\mathbb{E}\|\Psi^T M \Omega\|_F^4 \leq [q(q + 2)]^2 \|M\|_F^4.$$

Combining this with (3.5) by Cauchy-Schwarz, without assuming independence of R_0 and T , gives

$$\mathbb{E}\|R_0 T\|_F^2 \leq 6q\|[P_I, E]\|_F^2.$$

Minkowski and (3.8) therefore imply the useful signed-source estimate

$$\boxed{\sum_I \mathbb{E}\|(R_{0,I} - J_I(E))T_I\|_F^2 \leq 24qL\|E\|_F^2.} \quad (3.9)$$

The local constant used is $(\sqrt{6} + 1)^2 < 12$, followed by the commutator sum bound. This result is recorded with the local identities in `synchronized_union_route.md`. It prices the right separator and the source tail, but not the adaptive left lift.

3.3 The exact projected signed identity

For the completed synchronized rule define

$$\begin{aligned} Z_I^{\text{src}} &= (I - P)L_G(R_0 - J_I(E))T, \\ D_I^{\text{ret}} &= (I - P)L_G\Psi_I^T C_\perp, \\ H_I^{\text{inv}} &= (I - P)(I - Q_G)C_c. \end{aligned}$$

Then

$$(I - P)C_c = Z_I^{\text{src}} - D_I^{\text{ret}} + H_I^{\text{inv}}. \quad (3.10)$$

Indeed, left projection by $I - P$ annihilates the lifted retained query range, so $(I - P)L_G J_I = (I - P)L_G R_0$. Substitute (3.3), subtract the residual commutator, and restore the null component of C_c . The completion property (2.3) makes the first two terms visible and the last term invisible; hence these sectors are orthogonal. Since $C_\perp$ is physically orthogonal to W , the exact squared identity is

$$\|(I - WPW^T)C\|_F^2 = \|C_\perp\|_F^2 + \|Z_I^{\text{src}} - D_I^{\text{ret}}\|_F^2 + \|H_I^{\text{inv}}\|_F^2. \quad (3.11)$$

This formula preserves the sign and correlation of the source and return terms. Bounding them separately may destroy the cancellation which the formula is meant to expose. With arbitrary completion mixing visible and invisible coordinates, the omitted retained-query term would not in general vanish; the specified completion is therefore part of the identity's application to the learner.

3.4 The precise outstanding interface

Equation (3.9) does not imply a physical budget after left multiplication by $(I - P)L_G$. This lift, the tail R_0 , the separator, and the return term all share the training data. Products of their separate expectations are not legitimate substitutes for a joint estimate. Nor is $C_\perp$ fixed after conditioning only on the comparator. The invisible term remains relevant whenever the common predesign is singular, even if every retained child design is injective.

One sufficient all-input goal is a bound of order $L\|E\|_F^2$ for the sum of the actual outgoing and incoming comparator cut losses produced by (3.11), with the inherited defects accounted for without an additional depth factor. An equivalent or stronger direct bound on the final coefficient-space error would also suffice. The latter is the exact interface needed by the refit, rather than a scalar inverse trace alone.

To make the global requirement explicit, let $\mathcal{S}$ be the learned space. The sibling-block projection and its orthogonal row/column decomposition give the deterministic bound

$$\begin{aligned} \text{dist}_F(B, \mathcal{S})^2 &\leq \sum_{I \neq \text{root}} (\|(I - U_I U_I^T)B[I, I^c]\|_F^2 \\ &\quad + \|B[I^c, I](I - V_I V_I^T)\|_F^2). \end{aligned} \quad (3.12)$$

Each sibling block is charged to its own row and column endpoints; the displayed full exterior cuts only enlarge those nonnegative charges. Also

$$\text{dist}_F(A, \mathcal{S}) \leq \|E\|_F + \text{dist}_F(B, \mathcal{S}).$$

Thus a valid joint physical budget in (3.12) would imply (1.3). The unresolved point is establishing that budget along the actual adaptive recursion, not writing the final deterministic coefficient projection. No telescoping identity beyond the reviewed local statements is presumed here.

What is proved for independent source trees

4.1 The source interface is an assumption

The results in this chapter concern fixed physical source templates on disjoint supports. Fix all source physical data, and any allowed conditioning on an auxiliary stream Ω , *before* drawing the source blocks of Ψ . These blocks are independent standard Gaussian matrices. A source α has an orthonormal pretemplate W_α of width $2r$ and a fixed matrix target U_α with at most $s \leq r/2$ columns. Its original omission Gram is

$$C_\alpha = U_\alpha^T (I - W_\alpha W_\alpha^T) U_\alpha \succeq 0.$$

The represented coefficients may be arbitrarily large, and C_α may be singular or zero. Use $q = 8r$ shared query columns. The source response is $H_\alpha = \Psi_\alpha^T U_\alpha$. At a parent the pretemplate is the direct sum of the retained child frames, and the response is the sum of the compatible child responses. The full physical least-squares fit is retained before completion. Concretely, if $G = W^T \Psi_I$ is the current predesign and H_I its response, the fit is $F_I = W(G^\dagger)^T H_I$. In the additive ordered-prefix model, append the current template's coordinate vectors in their fixed order until the retained dimension is r , then recanonicalize the resulting projector in that same order. The parent template orders the left child before the right child. The fixed-guard model replaces this optional coordinate list by its specified pool.

In the additive equal-output model, the physical parent target is the direct sum of its children and $C_I = \sum_{\alpha \leq I} C_\alpha$. A more general compatible-transfer model allows deterministic matrices A_a, A_b with

$$U_I|_i = U_i A_i, \quad H_I = H_a A_a + H_b A_b, \quad C_I = A_a^T C_a A_a + A_b^T C_b A_b.$$

The guarded completion below handles strict drops in target spaces. Targets and transfers are inputs to this local interface; the theorem does not claim that the learner knows the physical target matrices.

There is a substantive difference between this assumption and a generic HSS residual. In an HSS construction, a child's response can contain exterior Gaussian anchors already exposed elsewhere in the computation. Arbitrary dense residual anchors need not have comparator-rank-sized span, and two independent arrays of added product noise do not make their adaptive subtrees independent. No decomposition theorem is asserted here which replaces those effects by independent fixed sources at total cost $O(\text{OPT}_k(A))$.

4.2 The earlier fixed-core inverse-optimal theorem

There is a proved small-rank theorem in a related local regression model which should not be conflated with the full-fit theorem below. At its physical leaves, fix rank- r frames U_α and source matrices E_α satisfying $U_\alpha^T E_\alpha = 0$. Put $\beta = \sum_\alpha \|E_\alpha\|_F^2$ and draw one Gaussian propagation

stream X . At a merge let $W = \text{diag}(U_a, U_b)$, $G = W^T X_I$, and $F = [E_a^{\text{out}}; E_b^{\text{out}}]$. An orthogonal split $[T, S]$ retains $U_I = WT$ and defines

$$B_I = S^T (G^\dagger)^T X_I^T F, \quad E_I^{\text{out}} = F - W S B_I, \quad c_I = (R_I^\dagger)^T X_I^T E_I^{\text{out}}, \quad R_I = U_I^T X_I.$$

Every selector must be subtree-local and invariant in physical coordinates under right rotations of X_I . Independent side information may be fixed first. These requirements are part of the theorem, even if the physical mandatory cores are fixed.

Prescribe nested orthonormal physical core frames Q_I^{phys} of ranks $s_I \leq k$, independently of X , and require the fixed leaf frames to contain their leaf cores. At ordinary nodes use the inverse-optimal rank- r completion containing $Q_I = W^T Q_I^{\text{phys}}$. The objective is the trace of the inverse retained query Gram. If $K = GG^T$ and $\phi_I = \text{tr}[(Q_I^T K Q_I)^{-1}]$, its optimizer obeys the deterministic additive bound

$$V_I \leq \min(V_a, V_b) + \phi_I. \quad (4.1)$$

A Schur complement reduces the optional-space optimization to the smallest generalized eigenvalues of the complementary inverse metric. Their eigenvalue comparison with K^{-1} and principal-submatrix interlacing prove (4.1).

Exact physical containment cancels the adaptive template in the mandatory Gram:

$$Q_I^T G G^T Q_I = (Q_I^{\text{phys}})^T X_I X_I^T Q_I^{\text{phys}}.$$

Thus this *mandatory* Gram is Gaussian, although the whole selected Gram is not. With $s = s_I$,

$$\mathbb{E} \phi_I = \frac{s}{q - s - 1}, \quad \mathbb{E} \phi_I^2 \leq \frac{s^2}{(q - s - 1)(q - s - 3)}.$$

For $r = 2k, q = 16k$, the local innovation-energy telescope and a Cauchy–Schwarz estimate for the dependent forced-core term give

$$\mathbb{E} \|E_{\text{root}}^{\text{out}}\|_F^2 + \mathbb{E} \|c_{\text{root}}\|_F^2 \leq \left(\frac{6694}{1001} + \frac{2\sqrt{3}}{77} \right) \beta < 6.733\beta. \quad (4.2)$$

When *every ordinary mandatory rank is exactly k* , the additional averaging envelope $V_I \leq \frac{1}{3} \text{tr } K^{-1} + \frac{2}{3} \phi_I$ improves the coefficient to

$$\frac{455 + 4\sqrt{3}}{153} < 3.020. \quad (4.3)$$

The final root need not use the optimizer. These constants belong to different core-rank patterns and cannot be interchanged.

The proof extends to adaptive cores contained in independently fixed physical envelopes of dimension $m < q - 3$, provided feasibility in the actual pretemplate and the same locality and equivariance hold. Inverse compression bounds the forced inverse by the inverse trace of the fixed envelope’s Gaussian Gram, giving first moment $m/(q - m - 1)$ and second moment at most $m^2/[(q - m - 1)(q - m - 3)]$. It does not prove that a general noisy HSS learner has such feasible low-dimensional envelopes. The theorem and envelope extension are reviewed in `audit_nonlinear.md`, with the detailed local recurrences in `hss_complete_proof_chain.tex`. This earlier result remains a valid restricted stability theorem; neither it nor the later fixed-source theorem is a learned-noisy-core guarantee for arbitrary HSS inputs.

4.3 An exact current-state latent covariance

Let R_I be the retained query design, F_I the full physical fitted target, and e_I the current query residual. The relevant law is a law of a restricted current state. One may retain physical bases and fits, retained and predesign Grams, and the prescribed functions of these summaries. One may not also condition on discarded residual Grams, old query-coordinate designs, old response Grams, or arbitrary cross-Grams involving forgotten designs. Those enlarged records can destroy the Wishart law used by the representation.

Under the permitted summary, the actual joint state has an equivalent representation with a positive semidefinite latent covariance T_I . Conditionally on it and the current design, the residual is centered matrix Gaussian with row covariance the projector onto $\ker R_I$ and column covariance T_I . For an additive merge let $S = T_a + T_b$; with transfers replace this by $S = A_a^T T_a A_a + A_b^T T_b A_b$. On the compact support of S , of rank t , the new covariance is

$$T_I = D_S \mathcal{B} D_S^T, \quad S = D_S D_S^T, \quad \mathcal{B} \sim \text{Beta}_t(3r, r/2), \quad (4.4)$$

with a fresh compact beta factor in this equivalent construction. In particular,

$$0 \preceq T_I \preceq S, \quad \mathbb{E}[T_I \mid S] = \frac{6}{7} S.$$

This is a distributional representation of the original shared-probe algorithm's permitted current state. It is not an instruction to resample Gaussian probes in the physical algorithm.

For completeness, the matrix identity behind the renewal is elementary. Let $A \sim W_t(6r, I)$ and $D \sim W_t(r, I)$ be independent, put $X = A + D$, and use the symmetric square root to set $\mathcal{B} = X^{-1/2} A X^{-1/2}$. The congruence Jacobian factors the joint density into a Wishart density for X and a beta density for $\mathcal{B}$, proving their independence. For independent copies with these laws,

$$X^{1/2} \mathcal{B} X^{1/2} \sim W_t(6r, I), \quad \mathcal{B}^{1/2} X \mathcal{B}^{1/2} \sim W_t(6r, I).$$

The second assertion follows because the two matrices have the same eigenvalues and both laws are orthogonally invariant. It does not interchange noncommuting square roots pointwise. Factoring S only on its support handles singular covariances, with $T_I = 0$ when $S = 0$. The complete joint-law proof and review are in `matrix_latent_budget.md` and `matrix_latent_budget_audit.md`.

4.4 A depth-independent theorem at sufficiently large retained rank

Theorem 4.1 (Fixed-source moment closure). *In the fixed, disjoint, additive matrix-target source model with the specified ordered-prefix completion, suppose $r \geq 4096$ and $q = 8r$. There is an absolute finite constant B_0 such that for every finite height, every node I , and every deterministic output vector v ,*

$$\left\| \|(F_I - U_I)v\|^2 \right\|_{L^2} \leq B_0 v^T C_I v. \quad (4.5)$$

The constants are independent of height, represented coefficients, source geometry, and nonzero covariance eigenvalues. The actual retained physical inverse has a uniformly bounded first moment.

Here the actual inverse is $V_I = \text{tr}[(R_I R_I^T)^{-1}]$, and “uniformly bounded” refers to its unconditional expectation under the fixed-source model, not a bound conditional on arbitrary fitted or discarded data. The theorem, its root review, and a separate independent audit are

recorded in `fitted_energy_bootstrap_route.md`, `fitted_energy_bootstrap_root_audit.md`, and `fitted_energy_bootstrap_audit.md`.

Finite moment ladder. For an integer p define $E_p(h)$ as the supremum of

$$\frac{\| (F_I - U_I)v \|^2 \|_{L^p}}{v^T C_I v}$$

over the stated model class at height h , all $r \geq 4096$, and all deterministic v with positive denominator. If the denominator is zero, every source is exact in that direction, and full-fit retention propagates exactness. Thus no undefined zero-budget directions are hidden in this normalization.

For an integer $p \geq 1$, a Gaussian vector g of covariance K obeys

$$\| \|a + g\|^2 \|_{L^p} \leq \|a\|^2 + (2p - 1) \operatorname{tr} K. \quad (4.6)$$

For a scalar coordinate, the coefficient of $a^{2p-2j} \sigma^{2j}$ in the Gaussian even moment, divided by $\binom{p}{j}$, is $\prod_{\ell=0}^{j-1} (2p - 2\ell - 1) \leq (2p - 1)^j$. Comparison with the binomial expansion proves the scalar inequality; Minkowski over physical coordinates proves (4.6) without requiring coordinate independence in the final step.

The enlarged-core captured-inverse estimate gives, on the compact current target support,

$$\| \|D_w^{-1}\|_{\text{op}} \|_{L^m} \leq \frac{2^{15}}{r}, \quad 1 \leq m \leq \frac{3r}{16}.$$

Its small-eigenvalue tail is bounded by $\frac{1}{2}(x/2^{-14})^{3r/8}$ for $0 < x \leq 2^{-14}$ after normalization by r . Integrating this tail yields the stated moment range. A deterministic enlarged core accounts for all retained directions, so there is no unexcited padding contribution omitted from the inverse.

For a parent direction v , set $c = v^T C_I v$ and $\ell = v^T (T_a + T_b) v / c$ in the additive model. The covariance trace of the physical Gaussian fit increment is bounded by $c \ell V_{\text{pre}}$. The inverse bridge and compact target whitening express the needed joint product in terms of the fixed-core inverse, captured inverse, and fitted energy. Hölder is applied to these quantities in their *same joint realization*; independence is not presumed. At orders $p = 2, 4, \dots, 128$, the available moments and beta decay yield

$$E_p(h+1) \leq E_p(h) + K_0 \rho^h (1 + E_{2p}(h)), \quad K_0 = 55,000,000, \quad \rho = \frac{7}{8}. \quad (4.7)$$

The source estimate is $E_p(0) \leq 2$ at the orders used. The coarse terminal theorem gives

$$E_{256}(h) \leq \frac{2567}{6} \left(\frac{5}{2} \right)^h. \quad (4.8)$$

It requires only unconditional inverse moments from its own valid orbit disintegration.

There are seven, rather than infinitely many, moment reductions. Put $\gamma = 5/2$, $p_j = 2^{j+1}$, and $\lambda_j = \gamma \rho^{7-j}$ for $1 \leq j \leq 7$. All these λ_j exceed one, while

$$\gamma \rho^7 = \frac{4117715}{4194304} < 1.$$

Starting with $B_7 = 2567/6$, choose successively

$$B_j = 2 + \frac{K_0}{1 - \rho} + \frac{K_0 B_{j+1}}{\lambda_j - 1}, \quad j = 6, \dots, 1.$$

Summing (4.7) proves $E_{p_j}(h) \leq B_j \lambda_j^h$. The final step gives the finite absolute bound

$$E_2(h) \leq B_0 := 2 + \frac{K_0}{1 - \rho} + \frac{K_0 B_1}{1 - \gamma \rho^7}.$$

The rank threshold guarantees every Hölder moment used in the seven steps. This proves (4.5). $\square$

In particular,

$$\mathbb{E}[(F_I - U_I)^T (F_I - U_I)] \preceq B_0 C_I, \quad \|\|F_I - U_I\|_F^2\|_{L^2} \leq B_0 \operatorname{tr} C_I.$$

One explicit inverse consequence is

$$\mathbb{E} V_I \leq \frac{8}{55} + 2^{27/2} \left(1 + \sqrt{B_0} + \frac{1}{\sqrt{7}}\right)^2.$$

These deliberately conservative constants prove uniformity, not a practical error prediction. The scalar model $r = 2, q = 16$ lacks the moment orders used here; the theorem does not cover it.

4.5 Fixed guards and compatible target drops

For deterministic compatible target transfers with $s_I \leq s_{\max} \leq r/2$, set $c = r - s_{\max}$. At each source define the guard as the first c columns of W_α and the pool as its first $2c$ columns. At a parent let

$$\text{Pool}_I = \text{diag}(\text{Guard}_a, \text{Guard}_b), \quad \text{Guard}_I = \text{first } c \text{ columns of the pool.}$$

These physical vectors are fixed before the source Gaussian queries. Begin with the full fitted target range, scan the pool in its fixed order until rank r is reached, and choose the frame with its guard first. Since the fit rank is at most $s_{\max}$ and the pool has at least r columns, the scan is feasible and retains the guard.

The key deterministic fact is stronger than feasibility. For a pool prefix C_j , put $\kappa(j) = \text{rank}[(I - C_j C_j^T) U_I]$. The iteration

$$j_0 = c, \quad j_{n+1} = r - \kappa(j_n)$$

increases to a fixed point $j_\star + \kappa_\star = r$. The full prefit is exact on each contained prefix, so the actual scan retains $C_{j_\star}$. The compact Gaussian capture theorem shows that the fit adds exactly $\kappa_\star$ directions outside it, almost surely. Hence the actual retained space is precisely the span of a deterministic enlarged core and a full compact fit; no arbitrary unexcited completion remains. The iteration is a proof device. The algorithm only tests observed fitted ranks while scanning its known pool.

The compact physical Gram outside this enlarged core dominates the transferred original omission Gram. It therefore supplies a common deterministic normalization for the inverse bridge. The matrix-beta law and the seven-step ladder retain their constants, extending Theorem 4.1 to this guarded compatible transfer model. The proof and independent review are in `fixed_guard_completion_route.md` and `fixed_guard_completion_audit.md`. A represented injection lying in the fixed pool is fitted exactly and does not create new latent error; arbitrary new unrepresented injections require the separate freshness conditions below.

4.6 Two scalar results which remain useful special cases

4.6.1 One original residual source and arbitrary clean siblings

A different scalar theorem uses fixed original rank-two leaf frames, query width sixteen, and an inverse-optimal completion containing the full scalar fitted target. There is exactly one original nonzero residual source, of squared norm β ; all sibling targets are arbitrary fixed vectors already represented in their original leaf frames. The tree may have unequal-depth siblings. For this rule,

$$\mathbb{E}[\|E_{\text{root}}\|^2 + \|c_{\text{root}}\|^2] \leq \frac{308110}{23023} \beta < 13.384 \beta. \quad (4.9)$$

Here E_{root} is the physical component of the target outside the retained space and c_{root} is its coefficient fit error inside that space. These orthogonal components together price the physical fitted-target error. The zero-budget case is exact.

The mechanism is a weighted inverse contraction along the unique residual-bearing path. With latent variance τ , inverse $V = \text{tr } K^{-1}$, and mandatory fitted-direction inverse $\phi = \|\alpha\|^2 / (\alpha^T K \alpha)$, let $H_t = \mathbb{E}[\tau_t \phi_t]$ and $A_t = \mathbb{E}[\tau_t V_t]$. The clean sibling has an exact fixed-target Gaussian query response; its radius is not replaced by the radius of a noisy fitted response. Integrating this Gaussian response and its conditional uniform optional line gives

$$\mathbb{E}\phi_{\text{new}} \leq \phi + \frac{38}{297}V + \frac{13}{11}v_b, \quad v_b \leq \frac{11}{49}.$$

The beta variance factor is $6/7$, and $A_t \leq H_t + (11/49)\mathbb{E}\tau_t$. Thus the weighted recurrence contracts with

$$\frac{6}{7} \left(1 + \frac{38}{297}\right) = \frac{670}{693} < 1.$$

Summing the reviewed energy recurrence gives (4.9). The exact integrations and audited constants are in `fitted_target_clean_siblings.md` and `fitted_target_constants.json`.

This theorem is not superseded by the large-rank theorem: it treats a small-rank, one-source regime with a concrete constant and a different completion rule. It is also not a decomposition of a nonlinear many-source run into independent one-source runs. The clean sibling's Gaussian radius is the special input which prevents that inference.

4.6.2 Source averaging preserves second moments within a factor

For the scalar source model with pretemplate width four, retained rank two, width sixteen, and inherited-core-plus-full-fit completion, let $\mathcal{Q}$ contain the retained query data of the original sources and their deterministic query transfers. The source physical dual lift A_α has metric $M_\alpha = A_\alpha^T A_\alpha$ and target cross vector $g_\alpha = A_\alpha^T U_\alpha$. Averaging the augmented source Gram

$$\begin{pmatrix} M_\alpha & g_\alpha \\ g_\alpha^T & \|U_\alpha\|^2 \end{pmatrix}$$

conditionally on its actual normalized query state and transporting it by the query-determined congruences gives exact conditional means of both the inverse and squared fitted error.

In particular, if V is the actual inverse trace and $Z = \|F - U\|^2$, write $\hat{V} = \mathbb{E}[V \mid \mathcal{Q}]$ and $\hat{Z} = \mathbb{E}[Z \mid \mathcal{Q}]$. At every finite height,

$$\hat{V}^2 \leq \mathbb{E}[V^2 \mid \mathcal{Q}] \leq 11520 \hat{V}^2, \quad (4.10)$$

$$\hat{Z}^2 \leq \mathbb{E}[Z^2 \mid \mathcal{Q}] \leq 11520 \hat{Z}^2. \quad (4.11)$$

The factor is paid once over the original sources, not once per merge. Extended expectations are allowed; these comparisons alone do not bound growth with height or establish finiteness.

The proof controls affine physical errors, including their target cross terms. At a positive-omission source, conditional on its observed core coordinate and moving query radius, the physical vector is separately affine in three independent blocks: the inverse core radius, the represented tail fit, and a standard Gaussian block. Their uniform affine fourth-to-second-moment constants are respectively 256, $9 + 4\sqrt{2} < 15$, and 3. Tensorization yields $768(9 + 4\sqrt{2}) < 11520$. Exact-source endpoints are checked directly; the zero-moving-fit branch uses its fixed-prefix chi-square degrees and is not obtained by a singular positive-noise limit. Conditional independence of the original sources and positive-semidefinite transport extend the source estimate to the whole tree without multiplying the constant at every level.

Thus $\|\hat{V}\|_2 \leq \|V\|_2 \leq 48\sqrt{5}\|\hat{V}\|_2$, and likewise for Z . These statements and their independent review are in `conditional_source_metric_comparison.md` and `conditional_source_metric_comparison_audit.md`; the initialization and exact error transport are recorded in `represented_source_query_metric.md` and `scalar_query_error_transport.md`. This result is a rigorous bridge between actual and conditionally averaged scalar models. It does not replace the missing arbitrary-HSS source interface.

Fresh innovations and bounded-stage correction

5.1 Freshness is a physical filtration condition

To add a fresh internal target Z_I , it is sufficient to reserve a deterministic physical subspace which has not contributed to any earlier query response, target response, or selection. More precisely, let D_{old} contain all physical directions exposed by the entire previous computation, assume the old record is measurable in $P_{D_{\text{old}}} \Psi$, and require

$$\text{range } Z_I \perp D_{\text{old}}, \quad \text{range } W_I \subseteq D_{\text{old}}.$$

Then $\Psi^T Z_I$, conditional on the complete old record, is centered matrix Gaussian with column covariance $Z_I^T Z_I$ and independent query coordinates. Node-private orthogonal reserved spaces are one way to enforce this condition. Merely calling a response a new source does not enforce it: ordinary parent physical coordinates have already been queried by their children.

The exact law, represented-pool injections, and the age-weighted covariance accounting are in `guarded_internal_injection_route.md` and `guarded_internal_injection_audit.md`. A source born at the current node has age zero. One cannot attach the decay factor of an old leaf source to such an injection simply because the current node has large height.

5.2 Pure fresh resets have uniform moments

Consider the fixed-guard model with $q = 8r$, target width at most $r/2$, and pure resets

$$A_a = A_b = 0, \quad U_I = Z_I, \quad C_I = Z_I^T Z_I.$$

The source targets are likewise perpendicular to their fixed source pretemplates, or zero. The new Z_I satisfies the reserved-space freshness condition. Earlier learned child spaces remain in use; only the current target is reset.

Let $h = \text{rank } C_I$, so the deterministic guard core has width $j = r - h$. Condition on the full state just before injection, including the current width- $2r$ predesign. Take an orthonormal query basis beginning with the core image and write its physical preimages as $[A_{\text{core}}, B_{\text{pre}}]$. There are $t = r + h$ complementary query coordinates. With

$$K = B_{\text{pre}}^T B_{\text{pre}}, \quad V_{\text{core}} = \|A_{\text{core}}\|_F^2, \quad \text{tr } K = V_{\text{pre}} - V_{\text{core}},$$

the whitened fresh response supplies a standard $t \times h$ Gaussian matrix X . Its column projector is independent of the old state, and full-fit retention plus $j + h = r$ gives the exact identity

$$V_I = V_{\text{core}} + \text{tr}(K P_X). \quad (5.1)$$

Hence, with $\alpha = h/(r + h) \leq 1/3$,

$$\mathbb{E}[V_I \mid \text{pre-injection state}] = (1 - \alpha)V_{\text{core}} + \alpha V_{\text{pre}}. \quad (5.2)$$

Whitening is only on the support of C_I ; the zero-covariance case is the fixed-core branch.

The fixed physical core has $\mathbb{E}V_{\text{core}} = (r - h)/(7r + h - 1)$. The independent child relative-frame law gives

$$\mathbb{E}V_{\text{pre}} = \frac{7r - 1}{6r - 1}(\mathbb{E}V_a + \mathbb{E}V_b), \quad \frac{7r - 1}{6r - 1} \leq \frac{13}{11} \quad (r \geq 2).$$

Substitution in (5.2) closes the induction $\mathbb{E}V_I \leq 1/4$. The physical fit error includes the omitted target orthogonally, so

$$\mathbb{E}[\|(F_I - Z_I)v\|^2 \mid \text{pre-injection state}] = (v^T C_I v)(1 + V_{\text{pre}}).$$

Consequently, for all heights,

$$\mathbb{E}V_I \leq \frac{1}{4}, \quad \mathbb{E}\|(F_I - Z_I)v\|^2 \leq \frac{35}{22}v^T C_I v \quad (r \geq 2). \quad (5.3)$$

For $r \geq 8$, the second moment of a diagonal entry of a Haar rank- h projector is $h(h+2)/(t(t+2))$. Minkowski, the fixed-core inverse moment, and the relative-frame L^2 bound similarly yield

$$\|V_I\|_2 \leq 2, \quad \|\|(F_I - Z_I)v\|^2\|_2 \leq (1 + 5\sqrt{3})v^T C_I v. \quad (5.4)$$

These are uniform theorems for fresh or zero resets, reviewed in `fresh_injection_moment_closure.md`. They do not establish the same convex contraction when a history-dependent inherited target is present.

5.3 What remains true with a noncentral fitted mean

Let $X = M + G \in \mathbb{R}^{t \times s}$, where M is fixed, G is standard Gaussian, and $s < t$. Put $\mathcal{U} = \text{range } M$ and $p = \dim \mathcal{U}$. Rotations fixing $\mathcal{U}$ give

$$\mathbb{E}P_X = A_{\mathcal{U}} \oplus bI_{\mathcal{U}^\perp}, \quad 0 \preceq A_{\mathcal{U}} \preceq I, \quad \text{tr } A_{\mathcal{U}} + (t - p)b = s.$$

The exact useful bounds are

$$\frac{s - p}{t - p} \leq b \leq \frac{s}{t}, \quad \text{tr}(K\mathbb{E}P_X) \leq \frac{s}{t} \text{tr } K + \left(1 - \frac{s}{t}\right) \text{tr}(KP_{\mathcal{U}}) \quad (K \succeq 0). \quad (5.5)$$

There is no general lower Loewner bound $(s/t)I$ on the mean subspace: a weak mean direction can have leverage below that value when another mean direction is strong.

For the upper bound on b , fix a unit vector perpendicular to the mean. Its Gaussian row has squared norm $R \sim \chi_s^2$. Conditional on that row, rotate the column coordinates and apply a Schur complement. The leverage is $R/(R + D)$, where conditionally D is noncentral chi-square with $t - s$ degrees of freedom. Its Laplace transform is at most the central one. Integrating the reciprocal denominator bounds the leverage mean by the central value s/t . The lower bound follows from the trace identity and $A_{\mathcal{U}} \preceq I$; the metric inequality follows by also using $A_{\mathcal{U}} \preceq I$.

For singular noise covariance, first retain its deterministic mean-only range of dimension τ , then work in its orthogonal quotient. If the noise has rank ℓ , the quotient projector has ℓ noisy columns in dimension $t - \tau$. Thus its central coefficient is $\ell/(t - \tau)$. The original fit has full column rank exactly when the mean-only and noisy ranks add to the required column rank. One must include the forced mean-only projector and any otherwise unexcited completion; they cannot be discarded by pretending the covariance is invertible.

Applied conditionally to a valid fresh guarded merge with inherited complementary mean M , this gives

$$\mathbb{E}[V_I \mid \text{old state}] \leq (1 - \alpha)V_{\text{mean}} + \alpha V_{\text{pre}}, \quad V_{\text{mean}} = V_{\text{core}} + \text{tr}(KP_{\text{range } M}), \quad \alpha \leq \frac{1}{3}, \quad (5.6)$$

under the stated compact full-fit/core rank condition. The singular full-rank version improves α to $\ell/(r + \ell)$. The new term V_{mean} is coupled to the inherited physical metric. It is not bounded by the pure-reset theorem or by independence of the fresh part alone. Equation (5.6) is therefore a one-step bridge, not a completed general mixed-injection theorem. The exact projector statements and audit are in `noncentral_projector_route.md` and `noncentral_projector_audit.md`.

5.4 An optional correction with only two new query batches

There is a separate, elementary bounded-stage module which can help when a pilot has a small *global low-rank residual tail*. It should not be confused with a general HSS pilot-quality theorem. Fix any known pilot M_0 , including its complete construction record, and put $R = A - M_0$. Draw a fresh $N \times \ell$ Gaussian matrix Ω_c , where $\ell = k + p$ and $p \geq 2$, and query

$$Y_c = A\Omega_c - M_0\Omega_c = R\Omega_c.$$

Let Q span its range, of dimension $d \leq \ell$. Freeze Q , then obtain the exact transpose products $A^T Q$ and form

$$K = R^T Q = A^T Q - M_0^T Q, \quad M_1 = M_0 + QK^T.$$

The second batch consists of exact adapted queries. It is not a claim that previously recycled noisy compressed products are independent fixed-input products. There are $\ell + d \leq 2(k + p)$ new products.

If $\tau_k(R)$ denotes the squared error of the best *global* rank- k approximation, the Gaussian range-finder calculation gives

$$\mathbb{E}[\|A - M_1\|_F^2 \mid M_0] \leq \left(1 + \frac{k}{p-1}\right) \tau_k(A - M_0). \quad (5.7)$$

Indeed, decompose the fresh Gaussian matrix in an SVD of R into top- k and trailing blocks Ω_1, Ω_2 . A candidate in the sampled range gives residual bound $\|\Sigma_2\|_F^2 + \|\Sigma_2\Omega_2\Omega_1^\dagger\|_F^2$. The blocks are independent and $\mathbb{E}\|\Omega_1^\dagger\|_F^2 = k/(p-1)$, proving the claim.

Taking $p = k + 1$ uses at most $4k + 2$ new products and gives factor two. It is a contraction by ρ if $\tau_k(A - M_0) \leq (\rho/2)\|A - M_0\|_F^2$. It gives the desired error order if a separate pilot theorem proves $\mathbb{E}\tau_k(A - M_0) \leq \beta L \text{OPT}_k(A)$. Neither hypothesis follows from HSS structure alone. For example, many disjoint unit rank-one sibling blocks form an HSS-rank-one matrix with many global unit singular values. A zero pilot can have zero HSS comparator error but a large global rank- k residual tail.

If M_0 is HSS-rank- r_0 , the global rank- ℓ correction has an HSS representation of rank at most $r_0 + \ell$. It supplies neither a free same-rank output nor an efficient best-HSS projection. The module and its exact hypotheses are recorded in `bounded_stage_compression_route.md`. It is an optional conditional repair, rather than a replacement for the synchronized learner's missing arbitrary-input physical budget.

Current numerical evidence and the stopping point

6.1 A reproducible test of the actual synchronized recursion

The current experiment implements the completed synchronized learner of Chapter 2, using the actual arrays $A\Omega, A^T\Psi$, common child unions, the full-row rank- $\leq R$ branch, and one joint visible/kernel completion. All node responses reuse the original probes. The full learned coefficient-space projection is computed densely as a diagnostic, with exact physical leaf diagonal blocks. No independent random refit and no final rank- k reduction are simulated.

The saved run has **624 cases from 80 independent probe pairs**. The main grid uses 16 pairs at each of

$$(N, k) = (64, 1), (128, 1), (256, 1), (128, 2), \quad L = 3, 4, 5, 3,$$

for 64 independent pairs. Each pair is reused across three exact families and two noisy patterns at three noise scales. This yields 192 exact controls and 384 noisy cases. A separate set of 16 pairs, each used at three scales, gives 48 clean-parent witness cases. Cases sharing probes are paired observations, not additional independent samples. Within a fixed cell, its mean and standard error use the 16 independent complete trees.

The script, raw records, report, and independent audit are `synchronized_union_numerical_check.py`, `synchronized_union_numerical_check.json`, `synchronized_union_numerical_check.md`, and `synchronized_union_numerical_audit.md`. The recorded invocation uses 16 seeds per configuration. The audit reconstructed 216 estimate blocks across 39 cells from the saved records and independently checked six small dense coefficient projections; it did not rerun the main grid.

6.1.1 Exact controls and a clean-parent witness

The exact controls include nested HSS-rank- k inputs, nested HSS-rank- $2k$ inputs testing rank slack, and deficient or zero cuts. The largest full coefficient-space squared error was $3.16200589 \times 10^{-23}$, and the largest normalized true-cut loss was $1.71934437 \times 10^{-22}$. The rank- $2k$ controls test Theorem 2.1; they are not declared to have zero nominal rank- k approximation error. All these matrices have zero physical-leaf diagonal blocks, so no arbitrary dense coarse-leaf block is being declared compatible with a finer rank bound.

The witness uses $N = 128, k = 1$ and a parent $I = [0, 64)$, with

$$u = (e_0 + e_{16})/\sqrt{2}, \quad p_1 = e_1, \quad p_2 = e_{17}, \quad b_1 = e_{32}, \quad b_2 = e_{33}, \quad f = e_{64},$$

and $A = B + \epsilon E$ for $B = uf^T$, $E = p_1 b_1^T + p_2 b_2^T$, $\epsilon \in \{.01, .1, 1\}$. This embeds the fixed matrix mechanism from the separate-view reachability example in the new learner's actual feedback recursion. It does not reuse or assume that earlier example's reached child frames or query values.

In all 48 cases the physical design Ψ_I was 64×32 of rank 32, and therefore noninjective on the full physical space. Each common pretemplate design was 8×32 of rank eight. The

largest child-template loss of the unit target was $3.84603468 \times 10^{-29}$ and the largest parent loss was $4.49944901 \times 10^{-29}$. The current residual commutator is structurally zero; its direct floating-point evaluation had norm at most $5.14307095 \times 10^{-16}$. These observations check the clean-parent mechanism at reached states, rather than replacing the compressed injectivity condition by full physical injectivity.

6.1.2 Fixed noisy inputs and diagnostic limits

The two noisy patterns have fixed HSS-rank- k comparators and balanced perturbation energy across split levels. One uses dense sibling blocks; the other reuses fixed global factors and level couplings with additional alternating rank-one components. Perturbations are normalized to Frobenius norm one before multiplication by $\epsilon \in \{.001, .03, .3\}$. Matrix-construction seeds are distinct from probe seeds. The denominator is the specified $\|A - B\|_F^2$, *not* a computed value of $\text{OPT}_k(A)$.

The largest cell mean of full coefficient-space squared error divided by this residual budget was

$$1.00792492 \pm 0.00335957$$

for $N = 256, k = 1$, the dense pattern, and $\epsilon = .3$; the uncertainty is one standard error. The largest individual ratio was 1.02872904. The largest cell mean after additional normalization by L was $0.30198110 \pm 0.00073533$, and the largest individual depth-normalized ratio was 0.30789474.

The high-rank internal union branch was exercised 10,752 times across the two orientations. There were 6,720 full-raw internal selections, 7,680 full-range leaf selections, 12,288 truncated leaf selections, and 1,248 uncompressed root selections. No design-rank failures were observed. The largest observed design condition number was 8.60370. No sample-SVD singular value lay within a factor 100 of the declared sample cutoff. This last diagnostic does not audit every singular value used by the separate lifted-union and coordinate-completion tolerances; those tolerances are explicitly recorded in the report.

These finite observations support exact/slack capture and the clean-parent repair, and show that the noisy high-rank branch was actually used. They are not an all-depth inverse-moment estimate, an all-input $O(L)$ stability theorem, or an OPT-relative rank- k output guarantee. Small finite-sample condition numbers do not establish a tail bound on the adaptive inverse.

6.2 A signed-identity check on already sampled cases

The separate identity diagnostic uses 12 cases already in the main grid: three configurations, two noise patterns, and two scales, all at the first prescribed seed. It reuses *three* probe pairs and does not create twelve new independent trees. At 400 reached oriented internal nodes, the largest normalized vector residual in (3.10) was 2.72093×10^{-15} , and the largest residual in the squared orthogonal split (3.11) was 1.71304×10^{-17} . The largest separator residual was 3.05673×10^{-15} .

For nonnegligible source and return terms, the realized fraction

$$\frac{2\langle Z_I^{\text{src}}, D_I^{\text{ret}} \rangle_F}{\|Z_I^{\text{src}}\|_F^2 + \|D_I^{\text{ret}}\|_F^2}$$

was positive at 263 of 400 nodes, with extrema 0.928404 and -0.912101 . Thus the reached trajectories exhibit both cancellation and amplification. The check validates the implemented algebra and the need to retain the cross term; it does not estimate a universal expectation of

that cross term. The records are `synchronized_signed_identity_check.py`, `synchronized_signed_identity_check.json`, and `synchronized_signed_identity_check.md`, reviewed in the same synchronized numerical audit.

6.3 Unreviewed calculations at the stopping point

The following observations were communicated while work was being stopped. They had not been saved as independently reviewed new notes. They are segregated here to preserve the state of the investigation, and are not used in any theorem or constant above.

6.3.1 A proposed global discard accounting

For nested physical projectors, let the proposed row discard projector at an internal nonroot node be

$$D_I^U = \text{diag}(P_a^U, P_b^U) - P_I^U,$$

and at a physical leaf use $D_I^U = I - P_I^U$. Define D_I^V similarly. All projectors act in their indicated node coordinates. With the uncompressed root convention, the working quantities were

$$\mathcal{N}_U = \sum_{I \neq \text{root}} \|D_I^U B[I, I^c]\|_F^2, \quad \mathcal{N}_V = \sum_{I \neq \text{root}} \|B[I^c, I] D_I^V\|_F^2.$$

The communicated candidate identity was

$$\text{dist}_F(B, \mathcal{S})^2 \stackrel{\text{unreviewed}}{=} \mathcal{N}_U + \mathcal{N}_V - \sum_{\substack{I, J \text{ ordered} \\ I, J \text{ incomparable}}} \|D_I^U B[I, J] D_J^V\|_F^2, \quad (6.1)$$

with proposed consequences $\max(\mathcal{N}_U, \mathcal{N}_V) \leq \text{dist}_F(B, \mathcal{S})^2 \leq \mathcal{N}_U + \mathcal{N}_V$. Its intended purpose was to charge each new physical discard once, rather than summing full inherited cut losses repeatedly. The exact partition and orthogonality accounting, including endpoints, still required a recorded audit at the stopping point. In particular, (6.1) is not being used to strengthen the proved sufficient interface (3.12).

6.3.2 Retained design rank versus stacked design rank

A second communicated observation concerns standard full $8k$ leaves. Their Gaussian physical designs have full row rank almost surely, so any retained R -dimensional leaf space has an injective design. At a parent, the stacked predesign contains a child design of rank R and therefore has rank at least R . Every mandatory lifted direction is visible. Scanning visible candidates before kernel candidates can then complete the retained space to rank R without using the kernel. This suggests an induction of retained-design injectivity for every fixed input, even if the stacked width- $2R$ predesign is singular.

This short rank argument was not independently audited at the stop. It must not be confused with the reviewed exact-family theorem, which proves full stacked rank, or with a lower bound on singular values. Retained rank R alone does not eliminate a comparator component invisible to a singular parent predesign. The associated bottom-leaf union-rank calculation was not available in a sufficiently recorded and reviewed form to include as a separate result.

6.4 Proof-status ledger

The surviving chain has a definite completed part and definite missing interfaces. The independent fixed-space refit gives its uniform constant 1607. The completed synchronized learner is explicit and has an exact rank-slack capture theorem, a clean-parent statement, a fixed-comparator separator, the $24qL$ unlifted source budget, and the projected signed physical identity. Its saved numerical checks test the actual shared-probe recursion with the stated finite scope.

The independent-source theorems prove genuine uniform local stability under their own assumptions: fixed nested physical cores at small rank, the one-residual-source scalar path, conditional source averaging with factor 11520, sufficiently large-rank fixed sources, fixed-guard compatible transfers, and pure fresh resets. The noncentral projector result supplies a valid mixed one-step inequality with an explicit inherited metric cost. None identifies arbitrary HSS residual history with the required independent source interface.

For the current synchronized route, the open mathematical requirement is a joint physical source/return/invisible budget sufficient for (1.3). Separately, the full target requires an efficient same-rank reduction on the original target tree. The query-only best-HSS projection argument does not settle that computational requirement. The observations in Section 6.3 remain provisional records rather than additions to the proved chain.

Source provenance

The following local records supply the result statements, their assumptions, and the internal audits cited in this manuscript. They are provenance for the present research record, not public bibliographic publications. The assembled LaTeX file is self-contained. Filenames below identify the original supporting records in the project directory.

- `audit_nonlinear.md`
- `audit_refit.md`
- `bounded_stage_compression_route.md`
- `conditional_source_metric_comparison.md`
- `conditional_source_metric_comparison_audit.md`
- `fitted_energy_bootstrap_audit.md`
- `fitted_energy_bootstrap_root_audit.md`
- `fitted_energy_bootstrap_route.md`
- `fitted_target_clean_siblings.md`
- `fitted_target_constants.json`
- `fixed_guard_completion_audit.md`
- `fixed_guard_completion_route.md`
- `fresh_injection_moment_closure.md`
- `guarded_internal_injection_audit.md`
- `guarded_internal_injection_route.md`
- `hss_complete_proof_chain.tex`
- `latest_primary_sources.md`
- `matrix_latent_budget.md`
- `matrix_latent_budget_audit.md`
- `noncentral_projector_audit.md`
- `noncentral_projector_route.md`
- `represented_source_query_metric.md`
- `scalar_query_error_transport.md`
- `synchronized_signed_identity_check.json`
- `synchronized_signed_identity_check.md`
- `synchronized_signed_identity_check.py`
- `synchronized_union_exact_audit.md`
- `synchronized_union_exact_route.md`
- `synchronized_union_numerical_audit.md`
- `synchronized_union_numerical_check.json`
- `synchronized_union_numerical_check.md`
- `synchronized_union_numerical_check.py`
- `synchronized_union_route.md`

Bibliography

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- [2] J. Levitt and P.-G. Martinsson. *Linear-complexity black-box randomized compression of rank-structured matrices*. SIAM Journal on Scientific Computing, 2024; arXiv:2205.02990v3. <https://arxiv.org/abs/2205.02990v3>.